

Amir Ali Farzin

PhD Student · Australian National University

amirali.farzin@anu.edu.au | [amirali78frz.github.io](https://github.com/amirali78frz) | [GitHub](#) | [Google Scholar](#) | [LinkedIn](#) | +61 468 583 866

Professional Summary

PhD candidate at the Australian National University working on optimisation methods for modern machine learning systems. My research develops zeroth-order and saddle-point algorithms for large-scale and nonconvex min-max problems, combining theoretical guarantees with efficient implementation. I am motivated to translate rigorous optimisation insights into scalable solutions for real-world learning and decision-making problems, and I am eager to bring this expertise to research efforts in optimisation, machine learning, and systems modelling.

Education

Ph.D. in Engineering and Computer Science

Australian National University (ANU)

Supervisor: Prof. Iman Shames and Dr. Philipp Braun

June 2023 – Present

Canberra, Australia

B.S. in Mechanical Engineering

Sharif University of Technology (SUT)

GPA: 17.41/20 (3.78/4)

2017 – 2021

Tehran, Iran

High-School Diploma

Alborz High-School

GPA: 19.86/20 (4/4)

2013 – 2017

Tehran, Iran

Professional Experience

Visiting Researcher

University of Melbourne

Jan 2026 – Dec 2026

Melbourne, Australia

- Conducting collaborative research on zeroth-order optimisation methods and min-max problems in machine learning.

Student Researcher

Australian National University

Supervisor: Prof. Iman Shames

June 2023 – Present

Canberra, Australia

- Dynamical systems perspective of optimisation algorithms: stability and convergence.

Peer Reviewer

International Conference on Machine Learning (ICML), Conference on Neural Information Processing Systems (NeurIPS), SIAM Journal on Control and optimization (SICON), Transactions on Control of Network Systems (TCNS), Transactions on Automatic Control (TAC), Control Systems Letters (L-CSS), Conference on Decision and Control (CDC), European Control Conference (ECC)

Selected Publications

- [1] A. A. Farzin, Y.-M. Pun, P. Braun, A. Lesage-Landry, Y. Diouane, and I. Shames, “Min-max optimisation for nonconvex-nonconcave functions using a random zeroth-order extragradient algorithm,” *Transactions on Machine Learning Research*, 2025.
- [2] A. A. Farzin, Y.-M. Pun, P. Braun, and I. Shames, “Properties of fixed points of generalised extra gradient methods applied to min-max problems,” *IEEE Control Systems Letters* (L-CSS), 2025.
- [3] A. A. Farzin, Y.-M. Pun, P. Braun, T. Summers, and I. Shames, “Solving the offline and online min-max problem of non-smooth submodular-concave functions: a zeroth-order approach,” *accepted at International Conference on Machine Learning (ICML)*, 2026.
- [4] A. A. Farzin, Y.-M. Pun, P. Braun, and I. Shames, “Minimisation of quasar-convex functions using random zeroth-order oracles,” *Transactions on Machine Learning Research*, 2026.
- [5] A. A. Farzin and I. Shames, “Minimisation of Polyak-Łojasiewicz functions using random zeroth-order oracles,” in *2024 European Control Conference*, pp. 3207–3212, IEEE.

Projects

Zeroth-Order Minimisation

- Minimisation of structured functions such as quasar-convex, submodular, and Polyak-Łojasiewicz functions using random zeroth-order oracles in constrained and unconstrained settings.
- Solving minimisation problems under sector condition using random zeroth-order oracles.
- Closing the zeroth-order first-order convergence gap via input to state stability.
- Building a novel deterministic variance-free zeroth-order framework based on exact cubature

Zeroth-Order Min–Max Optimisation

- Min-Max optimisation for nonconvex-nonconcave and possibly non-differentiable objective functions using a random zeroth-order extra gradient algorithm.
- Solving the Min-Max Problem of non-smooth Submodular-Concave Functions: A Zeroth-Order Approach.
- Developing a Python package for zeroth-order solvers using random oracles. The current version of the package is available on <https://github.com/amirali78frz/ZoSolvers.git> and <https://pypi.org/project/ZoSolvers/>.

Saddle Point Problems

- Properties of fixed points of generalised Extra Gradient methods applied to min-max problems.
- Building a framework with guarantees to analyse the discrete-time algorithms from the viewpoint of continuous-time dynamical systems.
- Analysing the state-of-the-art minimax algorithms from the perspective of convergence to saddle points.
- Designing an innovative continuous flow with guarantees of convergence only to the saddle points.

Raceline Optimisation

- Developing a Python package for raceline optimisation, incorporating new and state-of-the-art methods. The current version of the package is available on <https://github.com/amirali78frz/OptiLine.git>.
- Building a dataset for raceline optimisation and training a neural network taking track map information as input and efficiently output optimised racelines and velocity profiles with high accuracy.

Talks and Presentations

- University of Melbourne and University of Sydney, 2025 – Invited talk on min-max optimisation for nonconvex-nonconcave functions using a random zeroth-order extragradient algorithm.
- IEEE European Control Conference (ECC), Stockholm, Sweden, 2024 – Contributed talk on minimisation of Polyak-Łojasiewicz functions using random zeroth-order oracles.
- IEEE Conference on Decision and Control (CDC), Rio de Janeiro, Brazil, 2025 – Scheduled talk on properties of fixed points of generalised extragradient methods applied to min-max problems.
- Optimisation Days at UNSW, 2025 and University of Melbourne, 2026 – Invited talk on properties of fixed points of generalised extragradient methods applied to min-max problems and beyond.

Research Interests

- Numerical optimisation algorithms with applications in machine learning
- Robust optimisation and min-max methods
- Sequential decision-making and multi-agent systems
- Zeroth-order and derivative-free optimisation

Technical Skills

Languages: Python, MATLAB, C++ **Tools:** Simulink, SOLIDWORKS, Arduino, Proteus, Git

Machine Learning: PyTorch, TensorFlow, OpenCV, JAX

Teaching Experience

Teaching Assistant (TA), ANU and SUT Sep 2022 – Dec 2025

- Control Systems (Convener: Dr. Philipp Braun), ANU, Semester 2, 2024 and 2025
- System Dynamics (Convener: Dr. Igor Vladimirov), ANU, Semester 1, 2024
- Automatic Control (Convener: Prof. Aria Alasty), SUT, Semester 1, 2022

High-School Academic Counselor, Alborz High-School Aug 2017 – Aug 2021

- Advised over 400 students on academic planning and managing their educational pathways.

Honors & Awards

- Ranked 162nd out of 150,000+ participants (Top 0.1%) in Iran's National University Entrance Exam (2017).
- Selected to participate in the competitive Machine Learning Summer School (MLSS) 2026.
- Merit-based exemption from Iran's university entrance exam for master's degree at Sharif University of Technology.
- Received a full scholarship package for international Ph.D. candidates from the ANU School of Engineering.
- Ranked 5th out of 50+ teams in F1TENTH Sim Racing League at CDC 2024.
- Won the Student Travel Award to attend the Conference on Decision and Control (CDC) 2025.
- Awarded an Australian Mathematical Sciences Institute (AMSI) Travel Grant to attend AMSI Summer School 2025.
- Top 15% of SUT entrants (2017).
- Recognised as a Silver Reviewer in ICML 2026.

Languages

English: PTE Academic: 90/90, TOEFL: 106/120 **Persian:** Native